

OBC v1.2 Datasheet

Features

- Features an STM32G431RBT6 MCU
 - ARM Cortex-M4
 - 128 kBytes of Flash
 - LQFP-64 Package
- HSE: 24MHz. LSE: 32.768kHz
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General Description

The **datasheet** document class makes it easy to write great looking data sheets using the LaTeX typesetting system. It follows the classic style used by most manufacturers of electronic components.

You can download the document class from [latex-datasheet-template](#) GitHub repository. The repository includes this example datasheet as **example.tex** and the document class as **datasheet.cls**. You can build the PDF document using command `latexmk -pdf`.

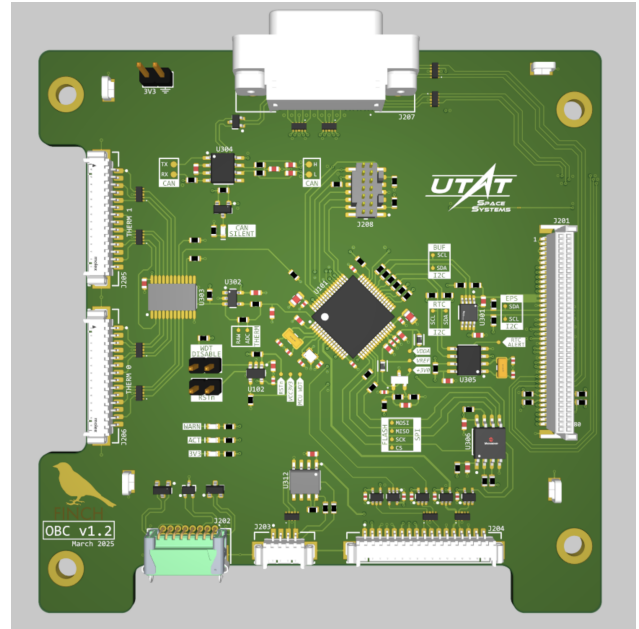


Figure 1: OBC v1.2

Block Diagram

Below is the block diagram that represents th high-level specification of the On-Board Computer.

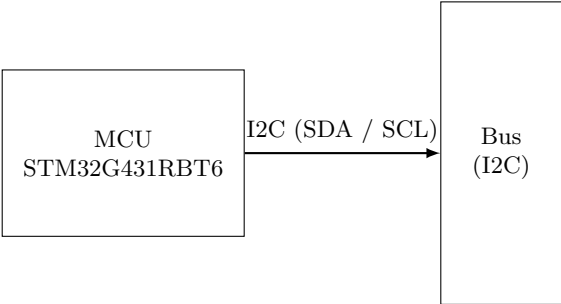


Figure 2: Block diagram: MCU and I2C bus

Bus

The table below maps the J201 connector pins found in the Protel netlist to their net labels.

Pin	Net
1	NetJ201__1
2	NetJ201__1
3	NetJ201__3
4	NetJ201__3
5	NetJ201__5
6	NetJ201__5
7	SRSVBUS
8	SRSVBUS
9	ADCSVCC5V2
10	ADCSVCC5V2
11	ADCSVCC3V3
12	ADCSVCC3V3
13	VCC3V3
14	VCC3V3
15	GND
16	EPS_I2C_B.SDA
17	EPS_I2C_B.SCL
18	GND
19	EPS_GPIO0
20	EPS_GPIO2
21	EPS_GPIO4
22	EPS_GPIO6
23	GND

Pin	Net
24	RF_RS422.PAY_RX_P
25	RF_RS422.PAY_RX_N
26	GND
27	STAR_RS485.STAR_RX_P
28	STAR_RS485.STAR_RX_N
29	GND
30	PAY_DEBUG.SWCLK
31	PAY_DEBUG.SWDIO
32	PAY_DEBUG.SWO
33	GND
34	BUS_GPIO0
35	BUS_GPIO2
36	BAT_SEL
37	GND
38	GSEVCC
39	GSEVCC
40	GSEVCC
41	GND
42	GND
43	GND
44	GND
45	GND
46	GND
47	GND
48	GND
49	GND
50	GND
51	GND
52	GND
53	GND
54	GND
55	GND
56	CANBUS.CAN_H
57	CANBUS.CAN_L
58	GND
59	EPS_GPIO1
60	EPS_GPIO3
61	EPS_GPIO5
62	EPS_GPIO7

Pin	Net
63	GND
64	RF_RS422.PAY_TX_N
65	RF_RS422.PAY_TX_P
66	GND
67	STAR_RS485.STAR_TX_N
68	STAR_RS485.STAR_TX_P
69	GND
70	PAY_DEBUG.UART_GSE_RX
71	PAY_DEBUG.UART_GSE_TX
72	PAY_DEBUG.RESET_n
73	GND
74	BUS_GPIO1
75	BUS_GPIO3
76	GNSS_PPS_B
77	GND
78	GND
79	GND
80	GND
MP1	GND
MP2	GND

RTC

Real time clock (RTC) is a battery-powered external clock available to OBC for timekeeping and synchronization purposes.

Electrical Specifications

Board Layout

Timing

Peripherals

Sensors

Thermistors

Manufacturing